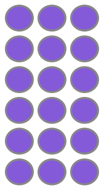
Expressing an integer as a product of its prime factors

Task A: Composite and prime numbers

1) Draw arrays to prove that the following are **composite** numbers.

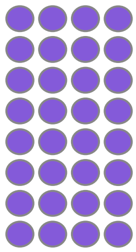
a) 18

e.g.

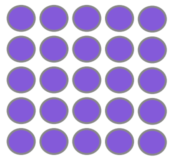


b) 32

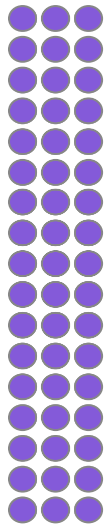
e.g.



c) 25



d) 51



Task B: Product of prime factors

1) Write the following as a product of their **prime factors**.

a) 78

e.g. $78 = 2 \times 39$

$= 2 \times 2 \times 13$

b) 165

e.g. $165 = 5 \times 33$

$= 5 \times 3 \times 11$

c) 525

e.g. $165 = 5 \times 33$

$= 5 \times 3 \times 11$

d) 840

e.g. $840 = 10 \times 84$

$= 2 \times 5 \times 2 \times 42$

$= 2 \times 5 \times 2 \times 6 \times 7$

$= 2 \times 5 \times 2 \times 2 \times 3 \times 7$

The working out steps are only examples, the final line should be the same, only the order can change.

Task C: Product of prime factors using index notation

1) Fill in the missing digits.

a) $2 \times 2 \times 2 \times 3 \times 3 \times 7 \times 7 = 2^{3} \times 3^{2} \times 7^{2}$

b) $2 \times 2 \times 5 \times 5 \times 5 \times 11 \times 11 = 2^{2} \times 5^{3} \times 11^{2}$

c) $5 \times 13 \times 5 \times 5 \times 3 \times 13 \times 3 \times 5 = 3^{2} \times 5^{4} \times 13^{2}$

d) $17 \times 2 \times 17 \times 7 \times 7 \times 2 \times 17 \times 7 \times 17 = 2^{2} \times 7^{3} \times 17^{4}$

e) $5 \times 2 \times 5 \times 7 \times 7 \times 11 \times 5 \times 7 \times 11 = 2 \times 5^{3} \times 7^{3} \times 11^{2}$

f) Why is there no box for the 2 in e?
The index would be 1 which we do not write.

2) Use $60 = 2^2 \times 3 \times 5$ to write the following as a product of their **prime factors**.

a) 120 $= 60 \times 2 = 2^2 \times 3 \times 5 \times 2 = 2^3 \times 3 \times 5$
b) 180 $= 60 \times 3 = 2^2 \times 3 \times 5 \times 3 = 2^2 \times 3^2 \times 5$
c) 300 $= 60 \times 5 = 2^2 \times 3 \times 5 \times 5 = 2^2 \times 3 \times 5^2$
d) 600 $= 60 \times 10 = 2^2 \times 3 \times 5 \times 2 \times 5 = 2^3 \times 3 \times 5^2$
e) 6000 $= 600 \times 10 = 2^3 \times 3 \times 5^2 \times 2 \times 5 = 2^4 \times 3 \times 5^3$
f) 4200 $= 60 \times 70 = 2^2 \times 3 \times 5 \times 7 \times 10 = 2^2 \times 3 \times 5 \times 7 \times 2 \times 5 = 2^3 \times 3 \times 5^2 \times 7$
g) 30 $= 60 \div 2 = 2^2 \times 3 \times 5 \div 2 = 2 \times 3 \times 5$